

PROFESSIONAL SUMMARY

Research scientist specializing at the intersection of machine learning and marine robotics. Backed by extensive hands-on experience, with a proven track record of translating complex algorithms into field-ready solutions and deploying physical robots in dynamic, real-world environments.

Core Expertise:

- **Machine Learning:** Bayesian learning theory, Gaussian & Sparse Gaussian processes, real-time applications.
- **Sensing & Navigation:** Multi-modal sensing (camera, sonar, radar), robust autonomous navigation systems.
- **Hardware & Deployment:** Real-world robotic deployment with resource-constrained edge devices.

RESEARCH EXPERIENCE

Postdoctoral Researcher, Texas A&M University

Oct 2024—Present

Computer Science Department

- **Algorithm Design:** Developing autonomous navigation algorithms for underwater robots operating under strict compute constraints, zero reliance on global localization, and minimal sensing.
- **Hardware Deployment:** Successfully optimizing and deploying computationally efficient methods onto physical robots utilizing edge devices, including Nvidia Jetson and Raspberry Pi platforms.

Doctoral Researcher, University of North Carolina at Charlotte

May 2020—Sep 2024

Bayesian Sensor Placement and Informative Path Planning

- **Software Engineering:** Developed efficient methods for sensor placement and informative path planning in continuous and discrete domains, notably developing the [SGP-Tools](#) Python library.
- **Theoretical Foundations:** Established guarantees for estimation uncertainty under motion constraints to ensure robust, predictable performance during autonomous data collection.
- **Performance Optimization:** Achieved processing speeds an order of magnitude faster than prior state-of-the-art solutions, enabling real-time deployment on resource-constrained field robots in marine environments.

Research Intern, GE Research

May 2023—Aug 2023

- **Explainable AI:** Created a low size, weight, and power-optimized Bayesian framework to explain and justify the predictions of pre-trained deep neural networks.

Graduate Researcher, University of North Carolina at Charlotte

Jan 2018—May 2020

Applied Deep Learning

- **Radar Pose Estimation (2019—2020):** Designed and evaluated deep learning models for human pose estimation and activity recognition using mmWave radar data.
- **Signal Processing (2018—2020):** Developed deep learning algorithms for gait-based user identification leveraging signals from WiFi routers and mmWave radars.

SELECTED PUBLICATIONS

- [1] **Kalvik Jakkala**, Saurav Agarwal, Jason O’Kane, and Srinivas Akella. “Informative Path Planning with Guaranteed Estimation Uncertainty”. In: *Robotics: Science and Systems (RSS)*. 2026. URL: <https://www.itskalvik.com/publication/uncertainty-guaranteed-ipp/>.
- [2] **Kalvik Jakkala**, Saurav Agarwal, Jason O’Kane, and Srinivas Akella. “Robot-Motion-Uncertainty-Aware Informative Path Planning with Guaranteed Estimation Uncertainty”. Under Review. 2026.
- [3] **Kalvik Jakkala** and Jason O’Kane. “Low-Cost Underwater In-Pipe Centering and Inspection Using a Minimal-Sensing Robot”. Under Review. 2026. URL: <https://www.itskalvik.com/publication/underwater-in-pipe-centering/>.
- [4] **Kalvik Jakkala**, Jason O’Kane, and Srinivas Akella. “Schur-MI: Fast Mutual Information for Robotic Information Gathering”. Under Review. 2026. URL: <https://www.itskalvik.com/publication/schur-mi/>.
- [5] **Kalvik Jakkala** and Srinivas Akella. “Fully Differentiable Adaptive Informative Path Planning”. In: *IEEE International Conference on Robotics and Automation, ICRA*. May 2025, pp. 5431–5437. URL: <https://www.itskalvik.com/publication/sgp-aipp/>.

[6] **Kalvik Jakkala** and Srinivas Akella. “Fully differentiable sensor placement and informative path planning”. In: *The International Journal of Robotics Research (IJRR)* (2025). URL: <https://www.itskalvik.com/publication/sgp-foundation/>.

[7] **Kalvik Jakkala** and Srinivas Akella. “Multi-Robot Informative Path Planning from Regression with Sparse Gaussian Processes”. In: *IEEE International Conference on Robotics and Automation, ICRA*. IEEE, 2024. URL: <https://www.itskalvik.com/publication/sgp-ipp/>.

[8] **Kalvik Jakkala** and Srinivas Akella. “Probabilistic Gas Leak Rate Estimation Using Submodular Function Maximization With Routing Constraints”. In: *IEEE Robotics and Automation Letters, RA-L* (2022). URL: <https://www.itskalvik.com/publication/graph-ipp/>.

EDUCATION

Ph.D. in Computer Science, University of North Carolina at Charlotte	Aug 2018—May 2024
• Concentration: Machine Learning and Robotics Advisor: Dr. Srinivas Akella • Thesis: Efficient Bayesian Sensor Placement and Informative Path Planning	
M.Sc. in Computer Science, University of North Carolina at Charlotte	Aug 2018—May 2021
• Concentration: Artificial Intelligence, Robotics, and Gaming GPA: 4.00 / 4.00	
B.Sc. in Computer Science, Wichita State University	Aug 2014—May 2018
• Minor: Mathematics GPA: 3.45 / 4.00	

TEACHING EXPERIENCE

Teaching Assistant, University of North Carolina at Charlotte	Jan 2021—May 2024
Mentored graduate students in: Machine Learning (ITCS 8156), Algorithms and Data Structures (ITCS 8114), and Optimization for Machine Learning and Data Science (ITCS 8010).	
B.S. Teaching Fellow, Wichita State University	Aug 2016—May 2018
Co-instructed and tutored undergraduate students in: Object-Oriented Programming (CS 311), Data Structures (CS 300), and Introductory C++ Programming (CS 211).	

SERVICE & AWARDS

• Associate Editor: IEEE International Conference on Robotics and Automation (ICRA)	Sep 2025—Present
• Co-organizer: IROS 2026 Workshop on Perception and Navigation for Marine Robotics	Sep 2026
• Reviewer: Top-tier robotics journals/conferences (ICRA, IROS, IJRR, RA-L, MRS, Ocean Eng.)	Jan 2022—Present
• Grant Recipient: UNC Charlotte Graduate School’s Summer Fellowship (GSSF) grant	May 2022
• Deans Honor Roll: Recognized for outstanding academic performance	2014—2018
• Vice President: Association for Computing Machinery (ACM)	Aug 2015—Dec 2016
• Vice President: Institute of Electrical and Electronics Engineering (IEEE)	Aug 2015—Dec 2016

SKILLS

- **Hardware & Edge Compute:** Nvidia Jetson, Raspberry Pi | **Sensors & Perception:** Sonar, mmWave Radar, Vision/Cameras
- **Programming Languages:** Python, C/C++, Matlab, Bash | **Libraries:** TensorFlow, PyTorch, OpenCV, ROS, GPflow, JAX
- **Platforms & Tools:** Linux, Docker, Slurm, Amazon Web Services (AWS), Microsoft Azure, Google Cloud Platform (GCP)
- **Domain Expertise:** Field Robotics Deployment (Autonomous Underwater/Surface/Ground/Aerial Vehicles)